Sampling and Intended Use

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach a sampling decision as a bounded evidence claim.

Replace copied audit tasks on the main route with complete condition-count examples while preserving earlier saved work.

Keep one habit: Assess the actual selected sampling plan; keep the old audit keys in review.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Define the population and conditions in an intended-use statement

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Use the full three-session route to compare an intended-use statement with complete fictional sample manifests. Teach condition counts and distinct plant IDs separately. Prior trip-inspection examples remain optional review. No model is fitted and no accuracy result is supplied.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Agreement records matching annotations, not correctness. Preserve rule version, source IDs and collection conditions so the sample can be compared with its intended use.

Printed lesson and workbook

Supplied sample manifests and cards for condition combinations and plant IDs

Optional Python 3 editor; no packages required

Blank coverage table: intended condition, observed photos, distinct plants, source, missing evidence, next action

Opening retrieval

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Write an observable label rule with boundaries and a review option

Reference example, not the only acceptable wording: Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table **schema**

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish **fitting** a rule from applying a fixed rule

Reference example, not the only acceptable wording: The **input** is width x and the label is the observed category 0 or 1. Candidates are 2, 4 and 6.

Selection uses mistakes on the four **training** cards only; held out labels must not guide it.

End goal

By the end, I can explain sampling and **intended use** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-4/5, ai-4/7, ai-4/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Label Quality

Python and Table Readiness

The ML Lifecycle

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

Concept to explain

Training examples should represent the people, objects, conditions, and devices described by the intended-use statement.

Term	Meaning
population	full group the product may encounter
sample	examples selected from that population
representation	coverage of relevant conditions

Term	Meaning
intended use	specific approved purpose and setting

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach coverage counts before choosing a sample

> B addresses named recorded-condition gaps; its model accuracy is unmeasured.

Complete current manifests

Sample	Plant IDs / condition	Photo count
A	P01–P09: day, camera A; one photo per plant	9
A	P10: evening, camera A	1
B	Q01–Q03: day, camera A	3

Sample	Plant IDs / condition	Photo count
B	Q04–Q06: day, camera B	3
B	Q07–Q09: evening, camera A	3
B	Q10–Q12: evening, camera B	3

Checked four-cell comparison and limits

Condition	Sample A	Sample B
Day/camera A	9	3
Day/camera B	0	3
Evening/camera A	1	3
Evening/camera B	0	3
Total photos / distinct plants	10 / 10	12 / 12

A is 9/10 daylight and contains no camera-B photos. B covers all four stated combinations with three photos each. That supports a narrower claim about these recorded conditions, not greater accuracy or full representativeness: species, labels, selection method and real intended-use frequencies are not supplied.

The program assumes valid nonempty tuples and correct condition strings; it counts records, not measurement truth. Different views of the same plant are related observations and must retain the shared plant ID for later split design.

Separate twelve-plant picture facts

Sample for the conditions in which the plant classifier will actually be used.

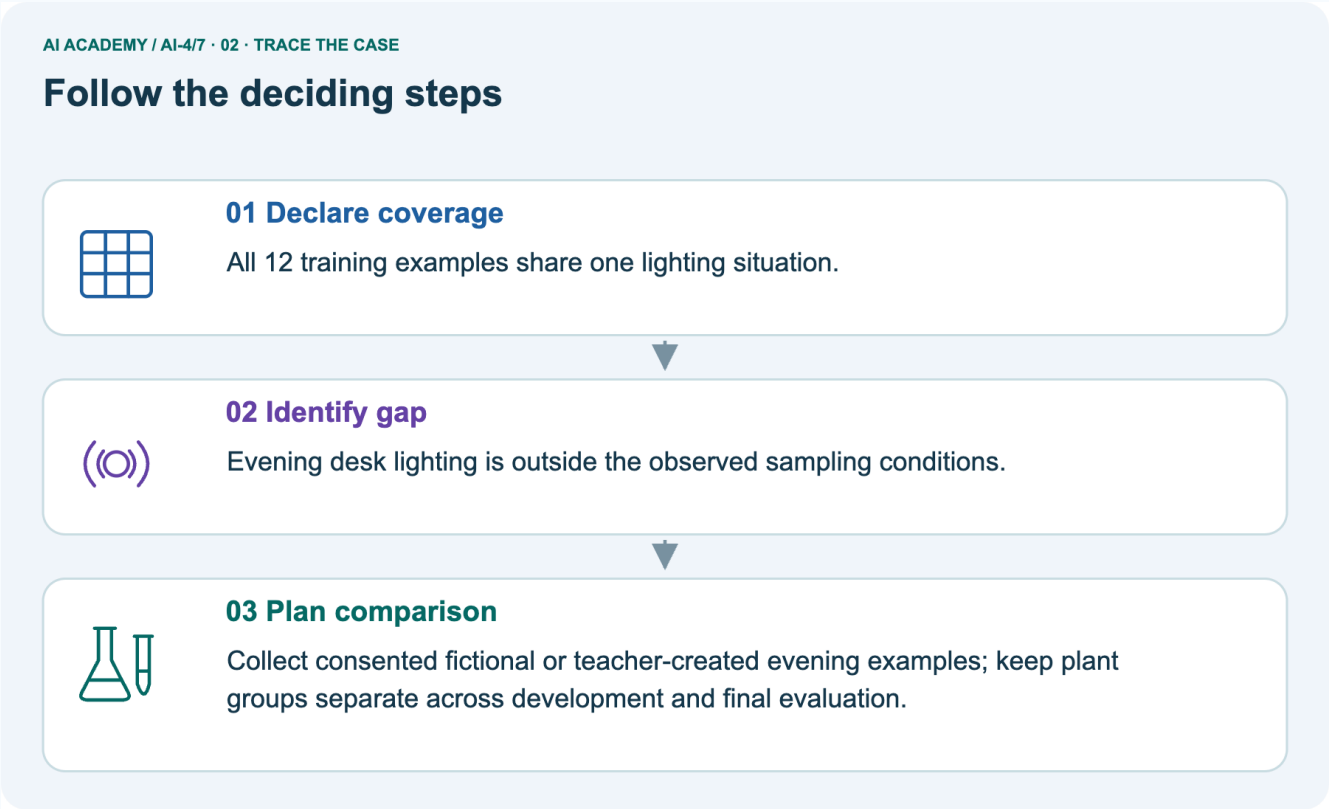
Training set: 12 windowsill plants photographed at midday. **Intended use** includes evening desk photos; no evening examples have been collected.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate twelve-plant example: all initial observations are midday windowsill photos and evening desk use is intended. Adding twelve new views of those same plants at midday gives24 photos but still12 distinct plants and no evening observations.

A collection plan should include the missing condition and record source groups; no exact performance change or guaranteed sufficient sample size follows from these counts.

Trace missing evening evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coverage	All 12 training examples share one lighting situation.
2. Identify gap	Evening desk lighting is outside the observed sampling conditions.
3. Plan comparison	Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

Predict same-plant midday additions

Change: Collect 12 more midday windowsill photos of the same plants.

Prediction: The evening-coverage gap remains.

Reason: More rows in the same condition do not supply **evidence** about the missing condition.

▼ Review · Earlier trip inspection

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

A successful run does not certify data quality.

Term	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero

Term	Meaning
Dataset card	A concise record of dataset contents creation intended use and limitations

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add **intended use**, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Compare two **sample**s against a deployment **population** and identify missing conditions.

MODEL IDEA **Training** examples should represent the people, objects, conditions, and devices described by the intended-use statement.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Intended use** includes indoor classroom plants under day and evening light. **Sample** A has 90% daylight photos from one camera.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 **Intended use** includes indoor classroom plants under day and evening light.

STEP 2 **Sample** A has 90% daylight photos from one camera.

STEP 3 **Sample** B balances lighting, camera distance, and several plant types.

STEP 4 Prefer B for this use, while documenting remaining gaps rather than claiming universal coverage.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Training photos use indoor white backgrounds; **intended use** is outdoors.

Identify the coverage gap.

Exact explanation to model

Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Intended Use

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide coverage reasoning and remove help

> Check each learner's counts before discussing preferred collection.

Current completion task and key

Using Coverage v1 A/B manifests, report day/A, day/B, evening/A and evening/B counts for each sample. Explain why B addresses a known gap without proving higher accuracy. If A photos are each copied ten times, state photo and distinct-plant counts and which cells remain missing.

A counts9,0,1,0; total10 distinct10. B counts3,3,3,3; total12 distinct12. B covers camera B and evening/B missing in A, but no model score, class coverage or real collection representativeness is shown. Ten copies per original A photo give100 photos and10 distinct plants; day/B and evening/B stay0.

Individual sampling misconception check

A teacher says100 copied indoor day photos prove reliable evening camera-B performance. What evidence is missing, and which counts can increase without closing that gap?

Copies do not supply evening/camera-B observations. Preserve ID groups and collect relevant evidence; no model performance has been measured.

Pictorial question and change

Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Collect suitable evening desk **observation**s, with plant grouping and labels recorded, rather than only repeating midday views. Changed-case check: The evening-coverage gap remains.

More rows in the same condition do not supply **evidence** about the missing condition.

Changed-case reasoning: The evening-coverage gap remains. More rows in the same condition do not supply **evidence** about the missing condition.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: **Training** photos use indoor white backgrounds; **intended use** is outdoors.

Write the question to explain: Identify the coverage gap.

Reveal after **prediction**s: Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Use the earlier transfer question as additional practice. Reserve the selected outdoor/other-camera collection plan for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: 2 of 5 raw rows have unknown duration, or 40 percent. No: the extra B occurrence is already among the missing rows, so subtracting all three flag counts double-counts it.

A distinct usable-trip count requires a stated resolution policy and source review. Exit examples include duplicate IDs, impossible values, units, **provenance** and coverage.

A defensible statement is that this five-row fictional export has documented quality flags and is suitable for inspection practice, with no model-readiness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve regression target meaning and why later outcomes are separate from prediction inputs. Establish that this week inspects data before fitting.
7 to 18 minutes	Read the source description and schema. Count physical rows and distinct trip IDs separately. Ask whether a repeated ID proves the reason for duplication.
18 to 32 minutes	Trace the code with row cards. Explain set, None and negative checks. Complete W1 and W2 with named denominators.
32 to 45 minutes	Use the guided question to show overlap: place both B cards in missing and repeated-ID groups. Do not subtract all flag counts independently.
Session 2 start to 8 minutes	Retrieve raw version versus corrected version. Ask what evidence could justify changing a recorded value.
8 to 20 minutes	Read the card example and complete W3. Distinguish a documented fact, an unknown and an assumption.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently. Require counts, overlap and a concise card with source and limitations before offering hints.
35 to 45 minutes	Discuss W5 and the exit claim. Use the fresh retry after support and record the support given.

Reduce help deliberately

Completion task: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Teacher check: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete intended-use coverage investigation

> Use condition counts as evidence about the sample, not measured model performance.

Current fixtures and checked guidance

A counts9,0,1,0; total10 distinct10. B counts3,3,3,3; total12 distinct12. B covers camera B and evening/B missing in A, but no model score, class coverage or real collection representativeness is shown. Ten copies per original A photo give100 photos and10 distinct plants; day/B and evening/B stay0.

Probes: B covers stated cells, one observation is sparse not guaranteed adequate, zero is documented missing coverage, outdoor camera C changes intended use. Independent plan must name missing outdoor/device conditions, collection provenance, labels and shared IDs, separate development/final evaluation and bounded slice conclusions; no score or guaranteed sample size.

Predict, count and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Sampling Audit: compare condition percentages, map them to **intended use**, and propose the smallest justified collection plan.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the evidence ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose sampling misunderstandings

MISCONCEPTION TO REPAIR A large dataset is representative simply because it contains many examples.

Current sampling misconceptions: more copied photos mean more independent plants—retain shared source IDs; nonzero coverage guarantees sufficient evidence—report sparse cells and uncertainty; B must be more accurate—no predictions are supplied; four filled cells prove all representativeness—species, labels and collection selection remain unknown; adding outdoor use requires no new evidence—revise the contract and collect relevant observations.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected outdoor/other-camera collection plan for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A complete card certifies data quality: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Missing means zero: None is unknown; zero is a specific allowed value in this exercise. Substitution without **evidence** changes the meaning.

Flag counts can always be added: The same row can have multiple flags. Count rows or unique events under an explicit policy and retain the raw denominator.

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected sampling plan

> The current workbook tests condition coverage, not old trip-audit arithmetic.

Selected task and checked answer

Selected case: **Training** plant photos are all indoors on one phone. **Intended use** includes outdoor photos on other cameras. Plan a sampling check before making a general accuracy claim.

Collect and evaluate relevant outdoor and camera conditions, recording source and condition coverage. Group related views of each plant when splitting.

More indoor photos alone do not establish outdoor performance; report the limits of measured slices.

Current W1–W3/W5 guidance

Current W1: four declared light/camera cells, rows versus distinct plants. W2: A counts9,0,1,0; total10 distinct10. B counts3,3,3,3; total12 distinct12. B covers camera B and evening/B missing in A, but no model score, class coverage or real collection representativeness is shown. Ten copies per original A photo give100 photos and10 distinct plants; day/B and evening/B stay0.

W3 rejects copied-volume performance claim. W5 names a zero cell, source/label/condition record and fresh source-separated evaluation. These replace copied audit exercises on the main route; their saved responses and earlier keys remain review.

Judge evidence and scope

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give a repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

▼ Review · Earlier audit questions and keys

W1 Define the row and source

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

W2 Interpret the audit counts

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

W3 Reject an unsupported correction

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

Those edits invent outcomes without source **evidence**. Preserve the raw version, flag the affected rows and investigate.

Record any supported correction with old value, new value, reason and version; unresolved values remain uncertain.

W4 Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the duplicate and missing flags overlap. Zero for T is allowed by the stated **schema**.

The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no **evidence** for real or outdoor navigation use.

W5 State an evidence gap

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

The row count increased but no new trip conditions or independent **observation**s were added. Coverage remains limited to the same supplied fictional indoor cases; the larger count does not establish broader usefulness.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Additional transfer practice question

Training photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

Reasoned practice answer

Reliable performance on indoor seedlings is unsupported. Collect representative, appropriately permitted examples from the intended conditions and evaluate those groups.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Large data volume guarantees coverage of every **intended use**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Evaluate outdoor lighting conditions and different devices, with reliable labels and adequate counts. More same-setting photos may improve indoor fit but leave the deployment shift untested.

Record background and object variation without treating every difference as the proven cause.

Explanations for the additional workbook examples

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use four condition cards and separate counters for photos and distinct source plants. Offer blank coverage headings rather than completed counts. Extend by asking which label, species and sampling-frequency dimensions the manifests do not establish.

Selected independent plan should cover outdoor lighting and new devices with documented sources/labels and shared plant IDs, separate development collection from fresh evaluation, and report per-slice limits without invented accuracy. Fresh retry: training only mature plants in daylight camera A; intended use seedlings under evening camera B. Require a condition-specific collection/evaluation plan and the distinction between unknown performance and proved failure.

Delayed return

Suggested return points in this level: ai-4/8, ai-4/10, ai-4/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 8: Leakage. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 8: Leakage. Carry forward source plant IDs, intended-use conditions and the distinction between development decisions and fresh evaluation evidence.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
</details>

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Inspect row meaning **missing value**s and repeated records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Reasoned practice answer: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Check every learner before discussion: Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Collect 12 more midday windowsill photos of the same plants. Require a **prediction** and a reason before comparing. Expected result: The evening-coverage gap remains.

Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

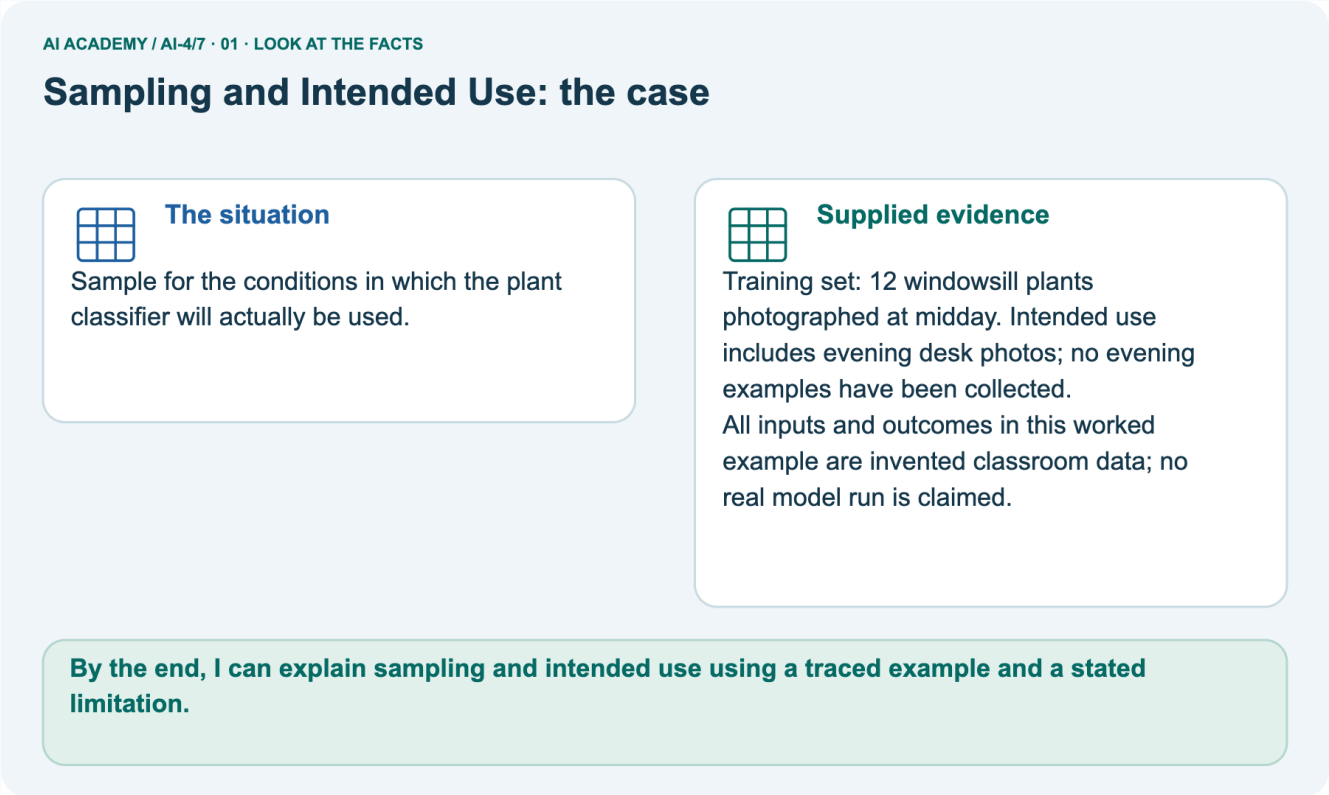
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Sampling and Intended Use: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

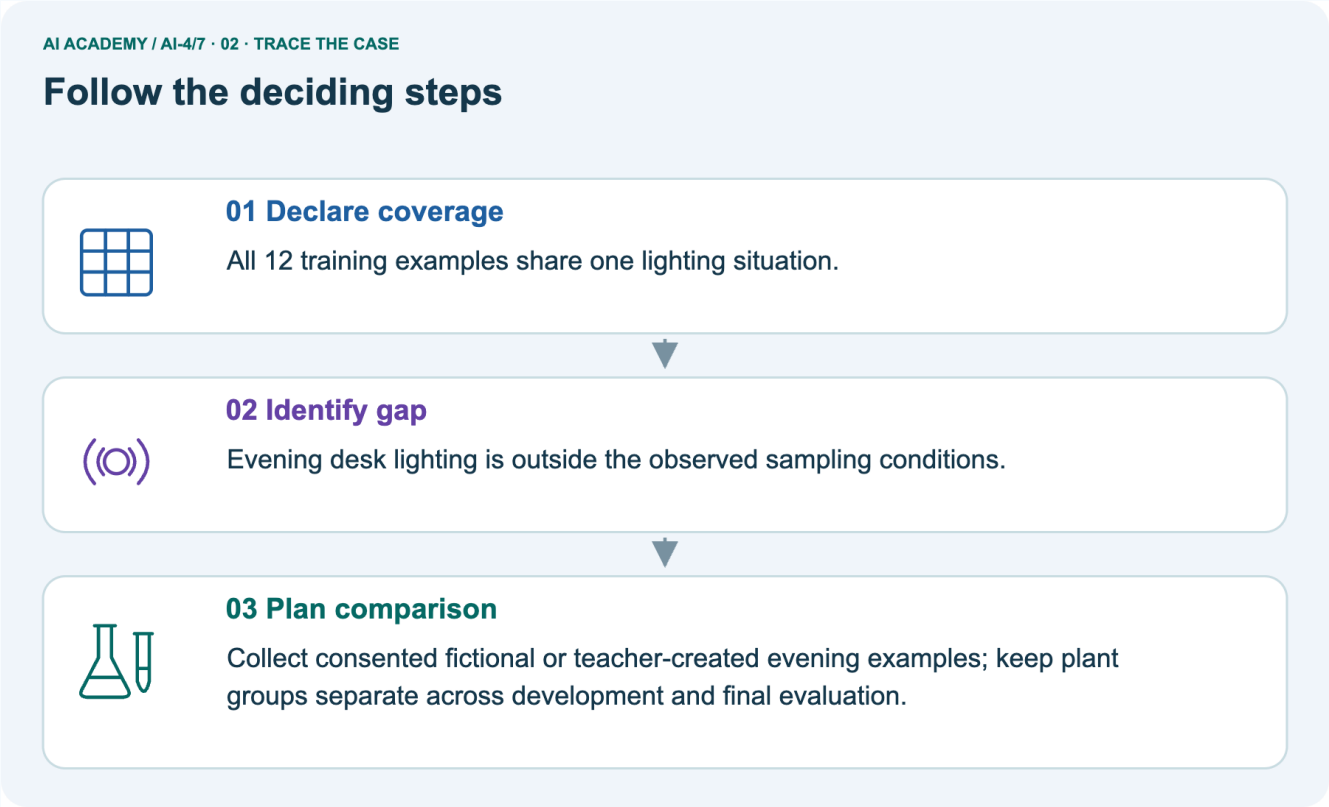
Sample for the conditions in which the plant classifier will actually be used. Supplied facts: **Training** set: 12 windowsill plants photographed at midday.

Intended use includes evening desk photos; no evening examples have been collected. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



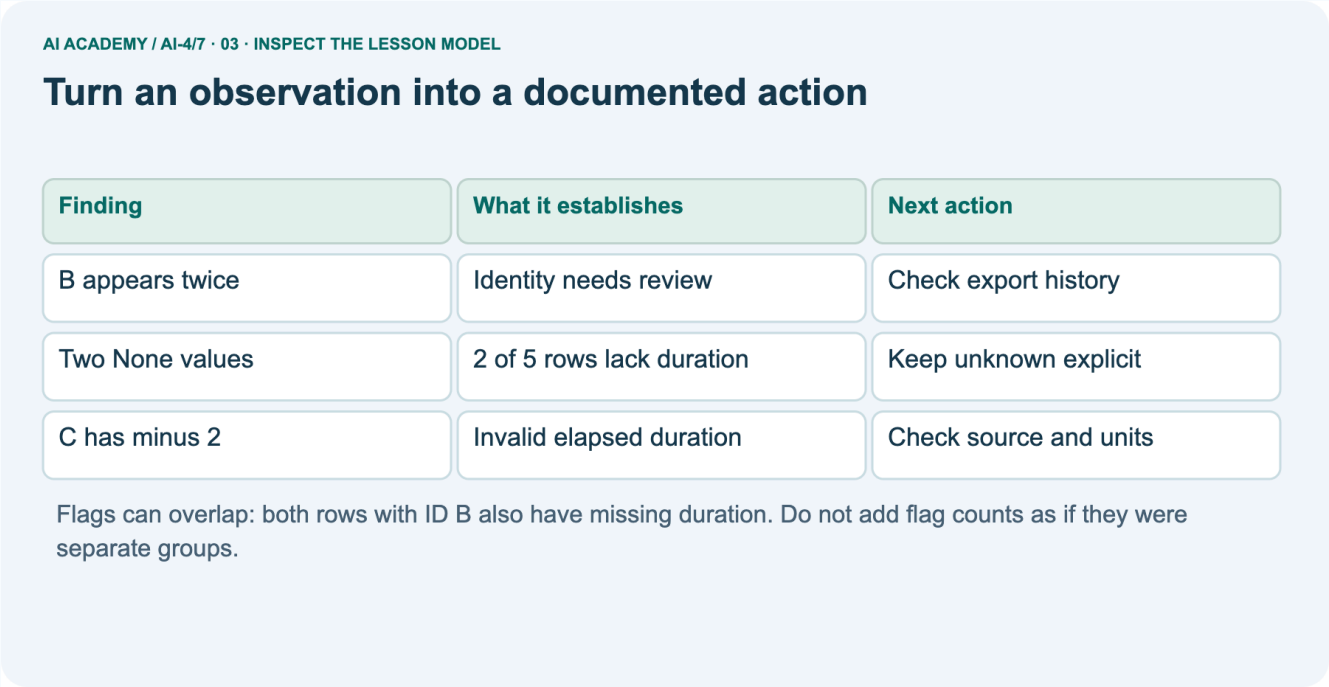
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coverage: All 12 **training** examples share one lighting situation. → Identify gap: Evening desk lighting is outside the observed sampling conditions. → Plan comparison: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

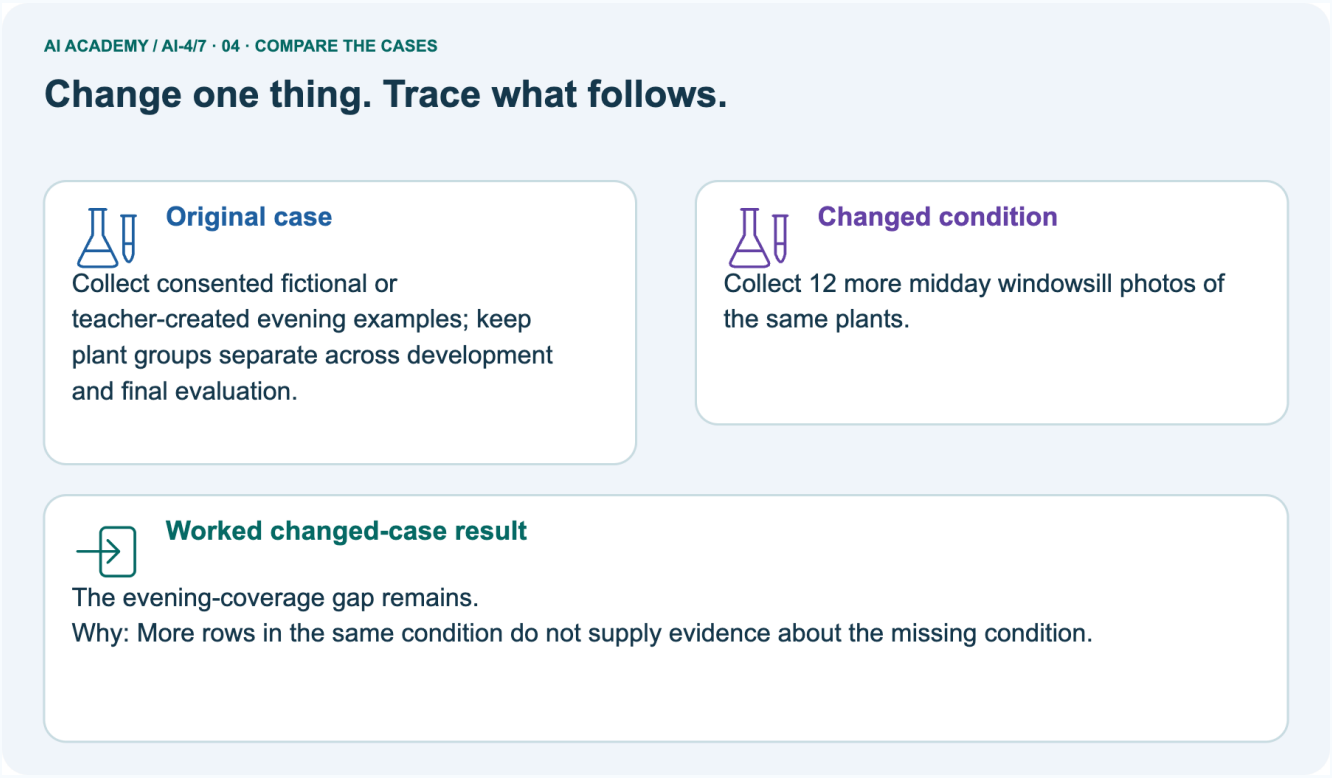
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

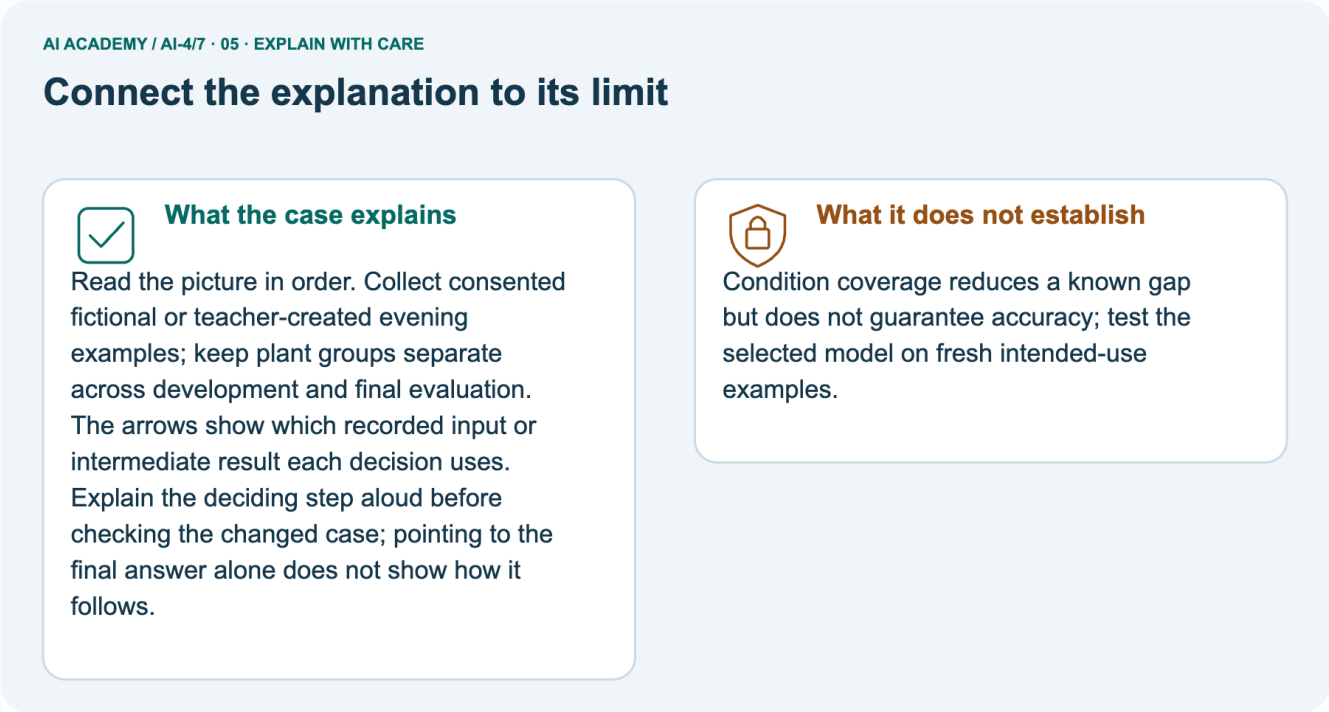
Changed condition: Collect 12 more midday windowsill photos of the same plants. Expected result: The evening-coverage gap remains. Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: More rows in the same condition do not supply **evidence** about the missing condition.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

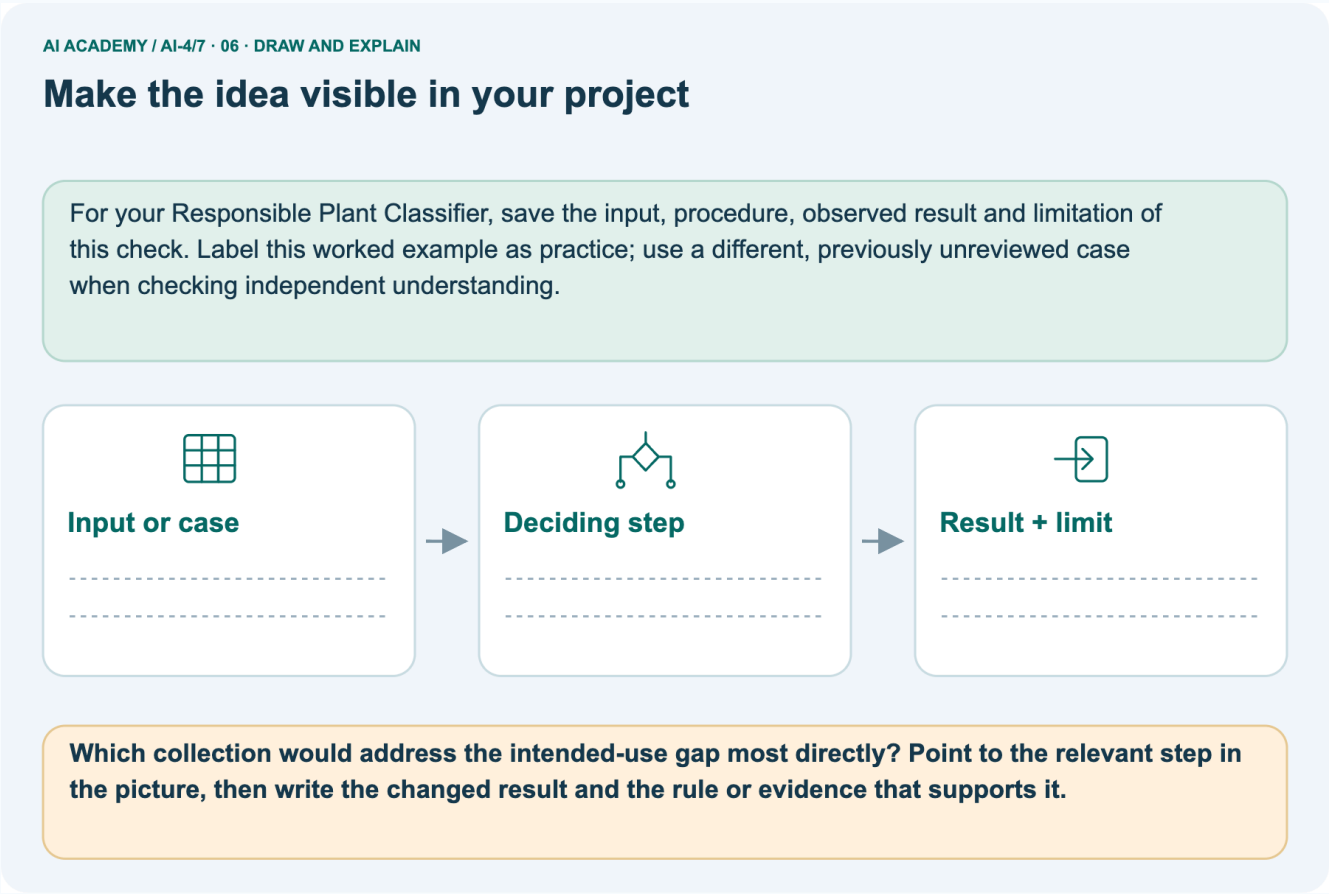
Limit: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Collect suitable evening desk **observation**s, with plant grouping and labels recorded, rather than only repeating midday views. Changed-case check: The evening-coverage gap remains.

More rows in the same condition do not supply **evidence** about the missing condition.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling and Intended Use: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.